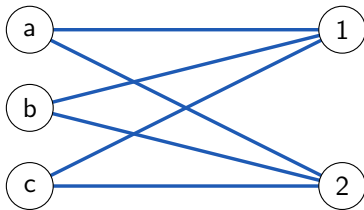


The Pigeonhole Principle

Section 2.4

Harris–Hirst–Mossinghoff, *Combinatorics and Graph Theory*



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Today's lecture

1. The Basic Principle

2. Choosing the Boxes

3. Number-Theoretic Examples

4. Monotone Subsequences

An existence principle

Question

If 13 people are in a room, must two have birthdays in the same month?

There are 13 people but only 12 months. At least one month receives two people.
We can force a coincidence without knowing which month it occurs in.

The pigeonhole principle

Pigeonhole principle

If more than n objects are placed in n boxes, then some box contains at least two objects.

Proof

If every box contained at most one object, the total number of objects would be at most n , a contradiction.

The generalized principle

Generalized pigeonhole principle

If N objects are placed in $k \geq 1$ boxes, some box contains at least

$$\left\lceil \frac{N}{k} \right\rceil$$

objects.

Equivalently, $k(r - 1) + 1$ objects force a box with at least r objects.

Proof of the generalized principle

Proof

Suppose every box has at most $\lceil N/k \rceil - 1$ objects. Since $\lceil N/k \rceil - 1 < N/k$, the total is less than

$$k \cdot \frac{N}{k} = N,$$

a contradiction.

Choosing the boxes

A useful strategy

Identify the desired coincidence, assign each object a label, and use the possible labels as boxes.

The hard part is usually finding the right labels.

Remainders, degrees, parity patterns, and lengths of subsequences will all serve as boxes.

Question

A drawer contains socks of five colors, with enough of each color. How many socks must be selected to guarantee four of one color?

Fifteen may give three of every color. Sixteen force four of some color:

$$5(4 - 1) + 1 = 16.$$

Equal degrees in a graph

Theorem

Every simple graph with at least two vertices has two vertices of the same degree.

Proof

For a graph of order n , degrees lie in $\{0, 1, \dots, n-1\}$. But degrees 0 and $n-1$ cannot both occur. Thus at most $n-1$ degree values are available for n vertices.

Remainders as boxes

Observation

Among any $m + 1$ integers, two have a difference divisible by m .

Proof

Place each integer in the box given by its remainder modulo m . Two integers lie in the same one of the m boxes, so their difference is divisible by m .

Theorem

For any integers $a_1, \dots, a_n$, some nonempty consecutive sum is divisible by n .

Introduce the $n + 1$ partial sums

$$s_0 = 0, \quad s_j = a_1 + \cdots + a_j \quad (1 \leq j \leq n).$$

Consecutive sums: proof

Proof

Two of $s_0, s_1, \dots, s_n$ have the same remainder modulo n . Say $s_i \equiv s_j \pmod{n}$ with $i < j$. Then

$$s_j - s_i = a_{i+1} + \dots + a_j \equiv 0 \pmod{n}.$$

Since $i < j$, the consecutive block is nonempty.

A divisibility problem

Question

Choose $n + 1$ distinct integers from $\{1, 2, \dots, 2n\}$. Prove that one of the chosen integers divides another.

Remainders alone are not the most useful labels here. Instead, remove all factors of 2 from each number.

Odd parts as boxes

Proof

Every positive integer has a unique form $2^r u$, where u is odd. The odd part u of a number at most $2n$ belongs to

$$\{1, 3, 5, \dots, 2n - 1\},$$

a set of n possibilities. Two chosen numbers have the same odd part, say $2^r u$ and $2^s u$ with $r < s$. The first divides the second.

Example

The n integers $n + 1, n + 2, \dots, 2n$ contain no distinct pair with one dividing the other.

Proof

If $a < b$ and $a \mid b$, then $b \geq 2a > 2n$, a contradiction.

Thus the threshold $n + 1$ in the preceding theorem is best possible.

Lattice points

Theorem

Among any five points with integer coordinates in the plane, two have a midpoint with integer coordinates.

Proof

Label (x, y) by the pair of parities of x and y . There are four labels. Two points share a label, so both coordinate sums are even. Their midpoint is an integer point.

Higher dimensions

Generalization

Among $2^d + 1$ points in $\mathbb{Z}^d$, two have a midpoint in $\mathbb{Z}^d$.

Proof

There are 2^d coordinate-parity patterns. Apply the pigeonhole principle.

The same labeling idea works without changing the proof.

A geometric application

Theorem

Among five points in a unit square, two are at distance at most $1/\sqrt{2}$.

Proof

Partition the square into four half-side squares, assigning boundary points consistently to one piece. Two points lie in the same piece. Their distance is at most the diagonal of that piece:

$$\sqrt{(1/2)^2 + (1/2)^2} = \frac{1}{\sqrt{2}}.$$

Approximation by rational numbers

Theorem

For every real number α and positive integer N , there exist integers p, q with $1 \leq q \leq N$ such that

$$\left| \alpha - \frac{p}{q} \right| < \frac{1}{qN}.$$

The boxes will be short intervals for fractional parts.

Rational approximation: proof

Proof

Place the $N + 1$ fractional parts of $0, \alpha, 2\alpha, \dots, N\alpha$ into the N half-open intervals

$$[j/N, (j+1)/N), \quad j = 0, \dots, N-1.$$

Two fractional parts differ by less than $1/N$. Subtract their corresponding multiples of α : for some $1 \leq q \leq N$ and integer p , we get $|q\alpha - p| < 1/N$. Divide by q .

Monotone subsequences

Subsequence

A subsequence keeps some terms of a sequence in their original order; the chosen terms need not be consecutive.

Example

In 3, 1, 4, 2, 5, the terms 1, 2, 5 form an increasing subsequence, and 3, 2 form a decreasing subsequence.

The Erdős–Szekeres theorem

Theorem

Every sequence of $(r - 1)(s - 1) + 1$ distinct real numbers has an increasing subsequence of length r or a decreasing subsequence of length s .

In particular, $n^2 + 1$ distinct numbers force a monotone subsequence of length $n + 1$.

Labeling each position

Proof

At position i , let l_i be the length of a longest increasing subsequence ending there, and let D_i be the corresponding decreasing length.

If the required subsequences do not exist, each label (l_i, D_i) lies in

$$\{1, \dots, r-1\} \times \{1, \dots, s-1\}.$$

There are only $(r-1)(s-1)$ possible labels.

Why the labels are distinct

Proof

For $i < j$, the terms are distinct. If $a_i < a_j$, append a_j to a longest increasing subsequence ending at i , giving $I_j > I_i$. If $a_i > a_j$, similarly $D_j > D_i$.

Thus no two positions have the same pair of labels. More positions than labels gives a contradiction.

A sharp example

Example

For $r = s = 4$, consider

3, 2, 1, 6, 5, 4, 9, 8, 7.

An increasing subsequence uses at most one term from each block, so has length at most 3.

A decreasing subsequence lies in one block, so also has length at most 3.

The general sharp construction

Use $r - 1$ decreasing blocks, each of length $s - 1$, and make every number in a later block larger than every number in an earlier block.

There are $(r - 1)(s - 1)$ terms, but no increasing subsequence of length r and no decreasing subsequence of length s .

Question

Show that among any 10 distinct integers chosen from $\{1, \dots, 18\}$, two have sum 19.

Find a partition into boxes that already encode the desired sum.

Proof

Use the nine boxes

$$\{1, 18\}, \{2, 17\}, \dots, \{9, 10\}.$$

Ten chosen integers force two in one box. Those two sum to 19.

- Too many objects for the available labels force a repeated label.
- Generalized pigeonhole bounds guarantee larger groups.
- Good labels include remainders, odd parts, and parity patterns.
- A pair of subsequence lengths proves the monotone subsequence theorem.

Inclusion and Exclusion